Manuscript outline, checked against the draft of 2026-10-10

enceladus_DSM `studies/keff_sintering/manuscript_outline.tex`

2026-10-10

Framing. Dry snow metamorphism is the story; the model covers one part of it, sintering by vapor transport, and the bulk conductivity that results, under conditions a laboratory can reach. Enceladus is the motivation and the closing implication. Every claim stays inside the simulated range: -40 to $-5\,^{\circ}\mathrm{C}$, air at 1 atm, sealed pores, 50 μm grains, 30 days, two dimensions.

How to read this. Read against `Literature/BaglinoJackson_DSM_manuscript.pdf` (52 pages). **[in draft]** the draft already covers it; **[missing]** not there yet; **[revise]** there, but it no longer matches the model or the results. Boxes mark where a figure or table goes. Themes are prompts, not text.

Where the draft stands

Section	State
Key points	empty
Abstract, plain-language summary	marked "needs to be updated"; both still claim temperature-dependent kinetics, "without retuning" and a collapse across porosity
1 Introduction	complete; kept as drafted (decision 2026-10-10)
2.1 Phase-field model	nearly complete
2.2 Gibbs–Thomson	**thin**: two equations; the clock, the thin-interface term and the α_c values are absent
2.3 Homogenization	complete
Numerical method; set-up of the aggregates	**no section**
3.1 Two-grain sintering	Molaro part written; saturated-pair paragraph is a placeholder
3.2 Aggregates, to Figure 6	written; the quantitative statements are absent
3.2 Aggregates, Figure 7	**weakest part**: stale text that contradicts the result
4 Discussion, 5 Conclusion	not yet reworked; 4.1 rests on a premise the model no longer has

Most in need of work, in order: (1) Section 3.2 from Figure 7 on; (2) Section 2.2; (3) the two missing Methods subsections; (4) the saturated-pair paragraph of 3.1.

Front matter

- **Title.** Kept (dry snow metamorphism is the paper's story); drop "Version 2" before submission.
- [missing] **Key points** (three). (i) For a given packing, temperature only rescales time. (ii) Conductivity follows specific surface area by one power law in well-connected packings; porosity sets the level. (iii) The model is the vapor route alone; what it does and does not reproduce of laboratory neck growth.
- [revise] **Abstract; Plain Language Summary.** Written last. Remove: kinetics "measured in experiments"; "without retuning"; "strong agreement" at $-5\,^{\circ}\mathrm{C}$; collapse "regardless of ... initial porosity".

1 Introduction

Decision (2026-10-10): the Introduction stays as drafted. The paper keeps the dry-snow-metamorphism story it began with, and concedes in the Methods that the model covers one part of it, sintering by vapor transport. The title and the heading of Section 2.1 therefore stay too. Only small fixes remain here:

- [revise] The last paragraph ends in a stray sentence fragment, says "28-day" (30), and "dependent of" (independent of).
- [revise] The same paragraph promises a predictor independent of "the initial porosity"; the result is narrower (Section 3.2).

Figure 1	[in draft] placed. Caption still says "neck of radius $R(t)$"; the figure shows $w(t)$.

Moved to the Methods instead (marked there): what the model leaves out; the definition of "desublimation"; why the comparison set is unloaded ice-sphere samples and not natural snow.

2 Methods

2.1 Phase-field model for dry snow metamorphism

- [in draft] Fields, Eqs. (1a)–(1c), ρ_{vs}^{I}, material laws.
- [missing] **Scope, stated up front** (moved here from the Introduction): of dry snow metamorphism the model carries sintering by vapor transport only. Not in it: surface diffusion, densification, overburden. Define "desublimation" here. One or two sentences on why the laboratory analogues are unloaded ice-sphere and ice-powder samples and not natural snow.
- [missing] M_{sub} and α_{sub} in terms of τ_{sub}, λ and ε; the temporal scaling ξ_T, ξ_v that the tables list but the equations do not show.
- [revise] Small things: "$T_0 = 273.15\,^{\circ}\mathrm{C}$" and "$T_0 = 271.15$ K" (both 273.15 K); "Allen–Kahn".

Table S1	`\TableMaterialProperties`. The draft has an *older* copy in the main text as its Table 1: it lacks the $\Delta\beta$ row and has $\tau_{\mathrm{sub}} = \varepsilon^2\beta_{\mathrm{sub}}/d_0$; its note points to two tables that are not in the draft ("Tables ?? and ??"). Re-import from `tables.tex`.

2.2 Interface kinetics and the Gibbs–Thomson condition

- [in draft] Eq. (6), the condition itself.
- [revise] **Eq. (7) has α_c in the numerator**; $\beta_{\mathrm{sub}} \propto 1/\alpha_c$. Symbols differ from the tables (d_{sub} for d_0, m_{mol} for m).
- [missing] The values: $\alpha_c = 1.020 \times 10^{-3}$ (aggregates), 0.1 (Molaro pairs), 1.002×10^{-3} (saturated pair), each with its β_{sub}.
- [missing] The thin-interface term $\Delta\beta$ and $\tau_{\mathrm{sub}} = \varepsilon^2(\beta_{\mathrm{sub}} + \Delta\beta)/d_0$, as in Kaempfer & Plapp (2009) and Moure & Fu (2024); what it costs (realised coefficient $\beta_{\mathrm{sub}} + \Delta\beta$).
- [missing] The sintering age $\theta = t/\tau_{\mathrm{sub}}$, defined here and not in passing in the Results; that ε is fixed across the study. *Open: state the clock as τ_{sub} at fixed ε, or as $\tau_R = \bar{R}^2\beta_{\mathrm{sub}}/d_0$.*

2.3 Homogenization for the effective thermal conductivity

- [in draft] Cell problem, weak form, indicator function, tensor law (16).
- [revise] One sentence breaks off ("commonly referred to as co"); "tonsorial".

Figure 2 [revise] Present, but it floats into Section 3. Move after Eq. (17). The figure now draws the 0.05 and 0.95 contours the caption names (2026-10-11); the lines are orange, the caption says "red". That band is $5.9\,\varepsilon$, 8.3 elements; the Karma width $2\sqrt{2}\,\varepsilon$ is 4. Add that only the x problem is shown.

2.4 Numerical implementation

- [missing] The whole subsection: discretization (quadratic C^1 splines), time stepping and the step cap, sampling of $\mathbf{K}_{\mathrm{eff}}$. Cite **Table S2** (`\TableKeffParametersFull`).
- [missing] Checks, one sentence each with a pointer to the supplement: domain size (Fig. S1), interface width (Fig. S2), time step and sampling (Fig. S3).

2.5 Aggregates: packings, boundary conditions, simulation matrix

- [revise] Now two sentences inside Section 3.2. Move here and expand.
- [in draft] Drop-and-roll deposition (named). [missing] void filling; five packings per porosity; acceptance tests.
- [missing] Why periodic boundaries: scale separation, no wall physics to invent, a closed system driven by curvature alone.
- [missing] The porosities and temperatures as lists; the master simulation used for single-run figures.

3 Results

[revise] **Opening paragraph.** Promises conductivity "independent of the mean temperature as well as the initial porosity" and a model that "captures sintering dynamics for varying temperatures". Restate as the three results.

3.1 Two-grain sintering

Unsaturated pair (Molaro et al., 2019).

- [in draft] Why this experiment; set-up; humidity inferred from the grain shrinkage; α_c at the upper end of the range; the authors' caveat on the $-5\,^{\circ}\mathrm{C}$ data.
- [missing] The numbers: 41% of the measured neck growth at $-20\,^{\circ}\mathrm{C}$; 170% at $-5\,^{\circ}\mathrm{C}$ (52.1 against 44.0 μm).
- [missing] That the $-20\,^{\circ}\mathrm{C}$ shrinkage is a calibration, so panels (c) and (d) are not a test at that temperature; neither the text nor the caption mentions those panels.
- [revise] "(cite other works as well)"; "table ??".

Table S3 `\TableGrainPairParameters`, cited where the draft has "table ??".

Figure 3 [in draft] placed. Caption stops at (b): add (c), (d).

Saturated pair (conditions of Demmenie et al., 2025).

- [missing] The paragraph (the draft has "Insert paragraph"). Content: equal grains, vapor saturated over the grains' own curvature; grain size holds (+0.06%); exponent 0.21 after relaxation (0.21–0.23 by window) against 0.26–0.33 measured and 1/3 in theory.
- *Open: interpretation. The Gibbs–Thomson prediction for this run is 0.29 and matches the final width to 4%; the run is faster early, where the neck fillet is about two interface widths across. Vapor transport alone gives 0.25–0.33 at Demmenie's own conditions for $\alpha_c \leq 0.1$. Parallel mechanisms add rates, so missing surface diffusion cannot raise an exponent.*

Figure 4 [in draft] placed, with caption.

3.2 Aggregates

Set-up.

- [in draft] 125 runs, periodic, 2 mm, 30 days. [revise] mesh "2829 × 2928" (2829 both ways).

Table 1 [missing] the parameter table (`\TableKeffParameters`). After the sentence that states the matrix.

What sintering does to the aggregate.

- [in draft] Description of the gallery.

Figure 5 [in draft] placed, with caption.

Temperature sets the clock.

- [in draft] Panels (b)–(d) described in words.
- [revise] Off-diagonal terms "at least one order of magnitude lower": k_{xy} reaches about 12% of k_{xx} on one packing; zero in the mean.
- [missing] The result stated: for every packing, k_{eff} at matched SSA agrees across the five temperatures within 0.27% (Fig. S4); rate follows τ_{sub} within 4%.

- [revise] "the trend is linear": linear in $\log\theta$, about 0.13 per decade, over a stated range.

> **Table 2** [missing] the temperature table (`\TableKeffTemperature`). After θ is introduced, before Figure 6.

> **Figure 6** [in draft] placed. [revise] Caption says "normalized by its initial value" (now by the value at $\theta_r = 30$) and $|\sigma| \le 0.31$ (the bar reads 0.34); define θ_r.

Conductivity as a function of the microstructural state.

- [revise] "3 packings for each porosity": five packings for each of three porosities.
- [revise] High porosities excluded because the ice "remains disconnected in the horizontal direction": not shown; what is shown is a weaker, scattered exponent and k_{xx}/k_{yy} of 0.79 and 0.64.
- [revise] Final paragraph ("28 days", "in ??", "no longer is dependent of porosity") contradicts the figure: porosity sets the level; only the exponent is shared.
- [missing] The numbers. Exponent -0.80; per porosity -0.79, -0.82 and -0.78. Level: $1.370\,\exp(-5.01\,\varphi)$, with 6.6% scatter between packings. Across porosity k_{eff} differs by a factor of about 3.3 at 30 d.
- [missing] The rise table by porosity and temperature; the two densest packings tie.

> **Figure 7** [in draft] placed. [revise] Caption for (a) says a single seed with seed scatter (it is the mean of five with their range) and stops at "(c)".

4 Discussion

State of the draft. Two subsections, written for the earlier model. Section 4.1 argues that $\beta_{\mathrm{sub}}(T)$ comes from Libbrecht's $\sigma_0(T)$, that nothing was retuned, and that $-5\,^{\circ}\mathrm{C}$ is "a genuine prediction". With a constant α_c none of that holds; the section has to be rebuilt, not edited. Section 4.2 claims a collapse across porosity, isotropy, and sintering "on the order of decades" near the Tiger Stripes. About 40 citations print as "?".

4.1 Temperature dependence of the kinetics

- Why the collapse holds: attachment-limited transfer, fixed α_c, one discretization; what would break it.
- Activation energy of the vapor route, about 50 $\mathrm{kJ\,mol^{-1}}$, against 24 measured by Choukroun et al.; the ranges overlap at 233–243 K.
- Reusable from the draft: the survey of how others set α_c (Bouvet, Braun) and the point that it is an effective parameter.

4.2 SSA as the state variable

- What the power law means; why porosity alone is not enough.
- Loss of the law at high porosity; anisotropy as a consistency check.

- Reusable from the draft: the comparison with Calonne et al. (density alone is incomplete; conductivity changes at constant density).

4.3 What the vapor route captures and what it lacks

- Figures 3 and 4 read together. Surface diffusion and the quasi-liquid layer; rate versus exponent.

4.4 What a laboratory could measure

- Time–temperature superposition and its activation energy; k_{eff} against SSA on unloaded ice-sphere samples; SSA against log time. Reusable from the draft: how SSA is measured.

4.5 Implications for plume deposits on Enceladus

- Qualitative only; no timescales in years. Reusable from the draft: the Choukroun and radar-attenuation paragraphs, without the "decades" claim.

4.6 Limitations

- Two dimensions; air-filled pores; one grain size; vapor route only; first day or so depends on the interface width; porosity ≤ 0.375. Reusable from the draft: conduction only; constant K_{ice}, K_{air}.

5 Conclusions

- **[revise]** Not yet reworked. Three results, each with its range of validity; the most useful experiment to do next.

Open Research; Acknowledgments

[missing] Code and data availability; run inventory (`RUN_TABLE.md`).

Supporting Information

- **Tables.** S1 material properties; S2 complete simulation parameters; S3 two-grain parameters.
- **Figures.** S1 domain-size convergence; S2 interface width; S3 time step and sampling; S4 temperature collapse for all packings; S5 porosity trends, SSA by porosity, anisotropy in time.
- **Text.** Packing construction; run table.